

Plasma etching of Si and SiO₂ in SF₆-O₂ mixtures

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The products of reaction and etch rates of Si and SiO₂ in SF₆-O₂ plasmas have been studied as a function of feed composition in an alumina tube reactor at 27 mHz, 45 W, and 1 Torr. There is a broad chemical analogy with CF₄-O₂ plasmas. As in CF₄-O₂ mixtures, the rate of Si etching and 703.7-nm emission from electronically excited F atoms each exhibit distinct maxima as a function of feed gas composition; these data support a model in which fluorine atoms, the etching species, compete with oxygen atoms for chemisorption on the Si surface. Without oxygen in the feed or Si in the reactor, no stable products could be detected. With an SF₆-O₂ mixture in the absence of silicon, the final reaction products are F₂, SOF₄, and SO₂F₂. The product distribution was unaffected by small SiO₂ substrates. When Si is etched, SiF₄ is the only stable silicon-containing etch product and SOF₂ is formed in oxygen-poor mixtures. Rapid etch rates ($\approx 10^4$ Å/min for Si) can be obtained with a high selectivity in favor of silicon (Si:SiO₂ > 40:1); thus SF₆-O₂ mixtures may represent an attractive alternative to CF₄-O₂ for the plasma etching of silicon and SiO₂.

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INTRODUCTION

Halogen-bearing gases have proved to be useful plasma discharge etchants for the manufacture of silicon integrated circuits. This has been an important motivation for recent studies of low-pressure radiofrequency discharges in freons and particularly in CF₄ and CF₄-O₂ mixtures.¹ These discharges are a source of fluorine atoms and active species, which gasify silicon and many of its compounds by forming the volatile product SiF₄.

There are very few published reports on plasma etching with inorganic fluorine-containing gases. Boyd and Tang^{2a} reported the use of SiF₄-O₂ as a practical etchant mixture, but only limited information^{1a,2} on the chemistry and reaction products is available. Heineke³ first showed that SF₆ could be used as a plasma etchant. Wagner and Brandt⁴ obtained mass spectra from the etching of Si by an SF₆ plasma and reported SiF₄ and S₂F₂ among the products of reaction; however, their results were inconclusive. Recently, Bruno *et al.*⁵ investigated the reaction of SF₆ in a quartz flow-tube reactor at relatively high pressure (20 Torr); the etching of the silicon dioxide walls of the reactor was measured and found to play an important role in determining the final product composition.

Others have studied the products of SF₆ decomposition in plasmas without intentional plasma-surface gasification.⁶ Several of these investigators found that SF₄, F₂, and F are formed,^{6c,d} while others reported^{4,6a} S₂F₂ and SF₂,^{6a} but the existence of SF₂ as a stable compound evidently is still in doubt.⁷ Interest in SF₆/H₂-fueled HF lasers led to exploratory characterization of the products of a plasma in this mixture⁸ (HF, SF₄), while Emeleus and Tittle^{6b} have analyzed the effluent from a microwave discharge in SF₆ mixed with Cl₂ and O₂.

The present study deals with SF₆ and SF₆/O₂ discharges in an alumina tube reactor at 1 Torr. The emission spectra from atomic fluorine and atomic oxygen were measured along with the product distribution. Si and SiO₂ etch rates were determined as a function of feed composition.

EXPERIMENTAL

The apparatus is outlined in Fig. 1; it is very similar to that described elsewhere.^{1g-h} The external capacitively coupled brass electrodes were driven by a 27-mHz rf generator and impedance-matching network. The water-cooled electrodes were 5 cm long, 5 cm in width, and machined to conform to the tube curvature with a fixed separation of 0.7 to 2.4 cm. The alumina reaction tube passivates toward reaction with F and F₂, giving a surface that does not promote the recombination of atomic fluorine⁹; the whole alumina surface was initially treated by sequentially moving the electrodes along the reactor axis to assure passivation. The mechanical pump was filled with Fomblin-Vac oil (Edwards) which is not attacked by the discharge products. All data were taken at 45 W, a pressure of 1.0 Torr, and a total flow rate of 27 cm³/min. The feed gas composition $x_{O_2} = [O_2]/([O_2] + [SF_6])$ was varied between 0 and 0.8.

Analysis of stable products was performed by infrared, gas chromatography (GC) and GC-mass-spectrometric techniques as detailed elsewhere.¹⁰

In order to avoid the hydrolysis of oxyfluoride compounds and SiF₄ by residual adsorbed moisture, the sampling bulbs were heated under vacuum prior to use and the gas-chromatographic column was pretreated with pure SiF₄; the GC carrier gas was dried with a liquid-nitrogen-cooled molecular sieve trap. SF₆, F₂, SiF₄, SOF₄, SO₂F₂, SOF₂, and O₂ were found in the effluent, while oxygen and fluorine atoms were measured by

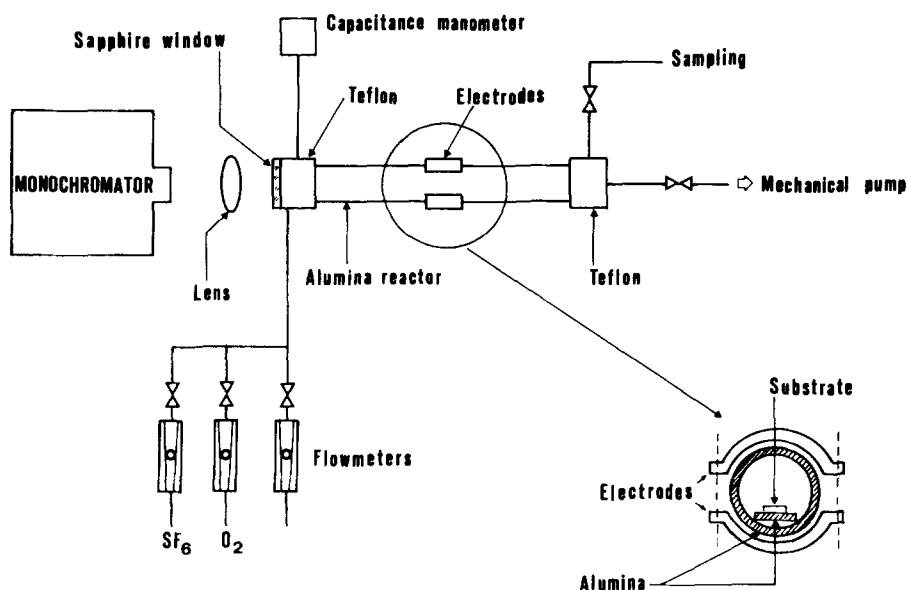


FIG. 1. Experimental apparatus.

monitoring the 777.2- ($3p^5-3s^5$) and 703.7-nm ($3p^2-3s^2$) emission lines with a 1-m Jarrel Ash monochromator. SO_2 was detectable, but never appeared as a product.

Commercially available UHP gases were used without further purification. (111) silicon and both fused and single-crystal quartz substrates were used to determine etch rates. During these experiments, the specimens were placed on a flat, passivated alumina substrate which faced the driven electrode.

Rapid etching was observed (see Figs. 5 and 6) which made it possible to determine both Si and SiO_2 etch rates gravimetrically (assuming uniform attack of the exposed surface). There was no significant difference between the etch rates of fused and single-crystal quartz. IR measurements of downstream SiF_4 concentrations during silicon exposure were in good agreement with those calculated from the etch rates and proportions of other products in the effluent. IR and GC sensitivities were too low to obtain adequate accuracy when SiO_2 was placed in the discharge.

RESULTS AND DISCUSSION

Discharge products

Within our detection limits, discharges in pure SF_6 without Si or SiO_2 substrates yielded no stable reaction products. While various investigators^{4,6} have reported SF_4 , atomic F, F_2 , S_2F_2 , and SF_2 among the products of SF_6 discharges, many of these studies were carried out in quartz discharge tubes where the tube wall participated in the reactions. Our findings are consistent with the observations of Bruno *et al.*⁵ in high-power, high-pressure discharges (20 Torr, 2–5 kW): without oxygen in the feed, etching of the quartz discharge tube produced SiF_4 , F_2 , and the sulfur-based carbonyl fluoride analogs SOF_2 , SOF_4 , and SO_2F_2 . They detected none of these products in an alumina tube without added oxygen.

The effect of added oxygen is shown in Figs. 2 and 3(a). In Fig. 2, the total conversion of SF_6 and oxygen to products is plotted against feed composition at 1 Torr

and a flow rate of 27 sccm. The product distribution is plotted against feed composition in Fig. 3(a). It is apparent that oxygen additions drastically increase the conversion of feed gas, evidently by reacting with fluorosulfur radicals and thus preventing their recombination with fluorine to reform SF_6 . This also leads to a net increase of F atoms (see Fig. 4). As the feed is made more O_2 rich, SO_2F_2 increases with respect to SOF_4 while the F-atom emission first increases, reaches a maximum, and then decreases (Fig. 4). These results closely parallel analogous trends in the carbon-based CF_4 - O_2 system^{1d,g-h} and suggest that SOF_4 is an initial oxidation product. Although SOF_2 formation has been reported at high power and pressure,⁵ in the absence of a Si substrate it was not detected under any conditions in the present study.

The product distribution was unchanged when SiO_2

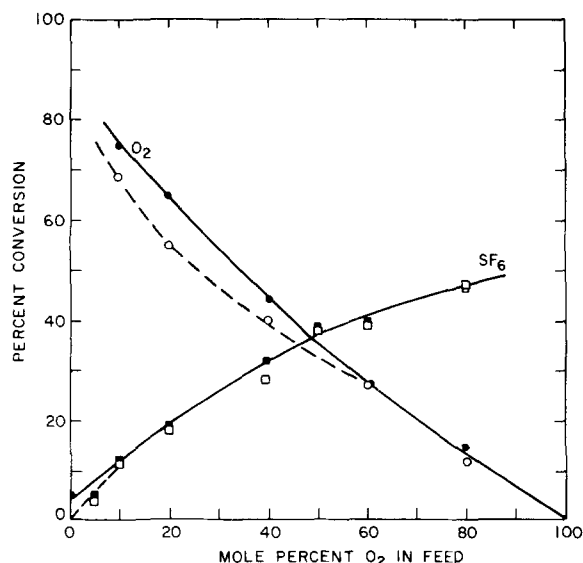


FIG. 2. Conversion of SF_6 and O_2 to products as a function of feed composition. SF_6 ($\square$) and O_2 ($\circ$) in empty reactor; ($\blacksquare$, $\bullet$) with a 0.3-cm^2 Si substrate present.

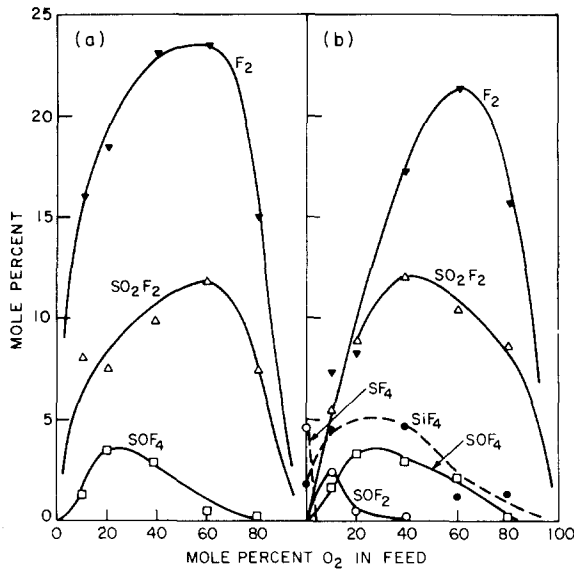


FIG. 3. Conversion of SF_6 into stable products in the absence (a) and presence (b) of 0.3-cm^2 substrates (vs feed composition).

samples (up to 3 cm^2) were placed in the discharge, but this is expected due to the small sample area and relatively low SiO_2 etch rate. When silicon is exposed in the discharge, a significant change in the product composition is observed [Fig. 3(b)]. SOF_2 is formed in oxygen-poor mixtures, SiF_4 appears, and the concentration of molecular fluorine is depressed. As the quantity of oxygen is increased, the etch rate goes through a maximum and subsequently decreases until it is too low to appreciably influence composition (Fig. 5).

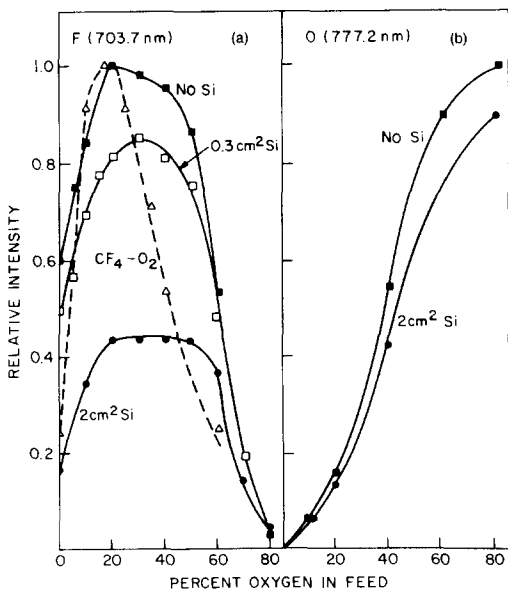


FIG. 4. Relative emission intensities from excited (a) F atoms and (b) O atoms vs feed composition: ■ empty reactor, □ in the presence of a 0.3-cm^2 Si substrate, ● in the presence of a 2.0-cm^2 Si substrate, and Δ emission from $\text{CF}_4\text{-O}_2$ normalized to the maximum F-atom emission intensity from $\text{SF}_6\text{-O}_2$.

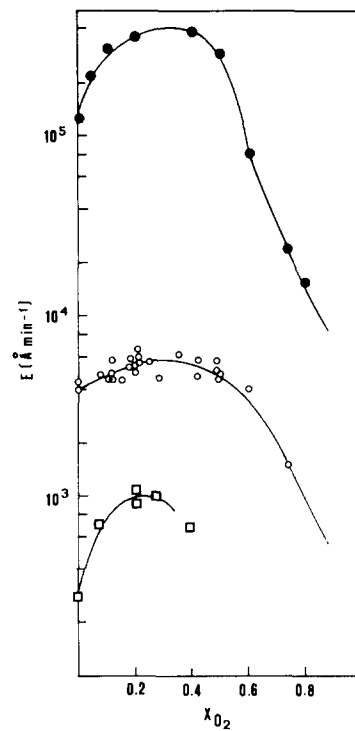
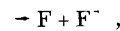
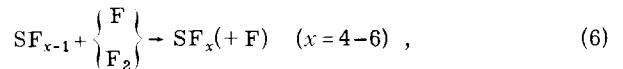
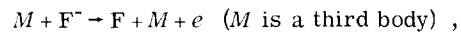
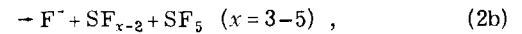
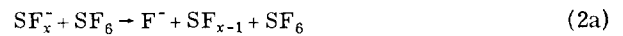
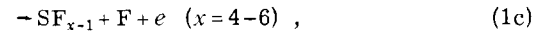
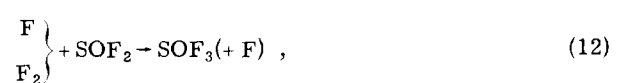


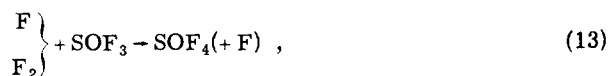
FIG. 5. Etch rates vs feed composition: ● Si in $\text{SF}_6\text{-O}_2$, ○ SiO_2 in $\text{SF}_6\text{-O}_2$, and □ SiO_2 in $\text{CF}_4\text{-O}_2$.

These results can be rationalized by the reaction sequence



in the absence of oxygen, and





where the reactions may be homogeneous (two or three body) or heterogeneous, depending upon the species involved.

In SF_6 , dissociative electron attachment [Eqs. (1a) and (1b), $x=6$] has an extremely large cross section at low energies and appears to be the main decomposition process.^{11a,b} Although appreciable SF_6^- is normally produced by dissociative attachment with near-zero-energy electrons at room temperature,^{11c} under plasma conditions the gas temperature is elevated and a moderate increase in temperature leads to a dramatic increase^{11d} in the cross section of reaction (1a) ($x=6$). A similar situation was noted¹² in SF_4 .

Some of the charge exchange reactions [Eq. (2)] have been reported,¹³ but collisional detachment may also produce SF_x radicals from the negative ions. Benson and Bott¹⁴ showed that the reversible sequence [Eq. (4)] accounts for experimental data on S_2F_{10} pyrolysis. In the discharge, electron impact dissociation [Eq. (5)] should prevail over the slow decomposition of S_2F_{10} to form¹⁴ SF_4 and SF_6 . Fluorine detachment [Eq. (3)], recombination [Eq. (7)], and dissociation reactions [Eq. (8)] have been studied in some detail.¹⁵

Wray and Feldman¹⁶ originally proposed Eqs. (11) and (15) to account for SO_2F_2 formation from $\text{SF}_6\text{-O}_2$ in a shock tube. By contrast, Siegel and Breisacher¹⁷ suggested that Eq. (10b) produces SOF_2 as an initial product during exploding-wire oxidation. However, the relative free energy¹⁸ of reaction favors (10a) ($\Delta G_{f,298} = -132$ kcal) over (10b) ($\Delta G_{f,298} = -13$ kcal).

Frost *et al.*^{6c} found that F and SF_4 are the main products extracted from the plasma zone of a SF_6 discharge. This is reminiscent of exploding-wire studies without Si or SiO_2 in which SF_4 and F_2 are the only detectable products.¹⁷ In general, SF_4 appears as a significant final product whenever fluorine is removed by the formation of a stable compound as, for example, by HF production with hydrogen additions⁸ or during SiF_4 production in the present experiments.

In oxygen-rich mixtures, reactions (11), (14), and (15) are favored over reactions (12) and (13) so that the ratio of SO_2F_2 to SOF_4 increases. When Si and small amounts of O_2 are present, F_2 is depleted by the formation of SiF_4 (Fig. 3), reactions (6), (12), and (13) are suppressed, and SOF_2 appears as a final product. As more oxygen is added to this mixture, the SOF_2 is rapidly oxidized (Eq. (15) and SO_2F_2 is favored over SOF_4 [Eqs. (16) and (14)].

The relative intensity of emissions from electronically excited F atoms (703.7 nm) and O-atoms (777.2 nm) are shown in Fig. 4(a). Assuming proportionality between the emission and concentration of atoms, it is apparent that the concentration of F atoms decreases as the area of Si in the discharge zone is increased (from 0–2 cm^2). Si substrates also produced a small decrease

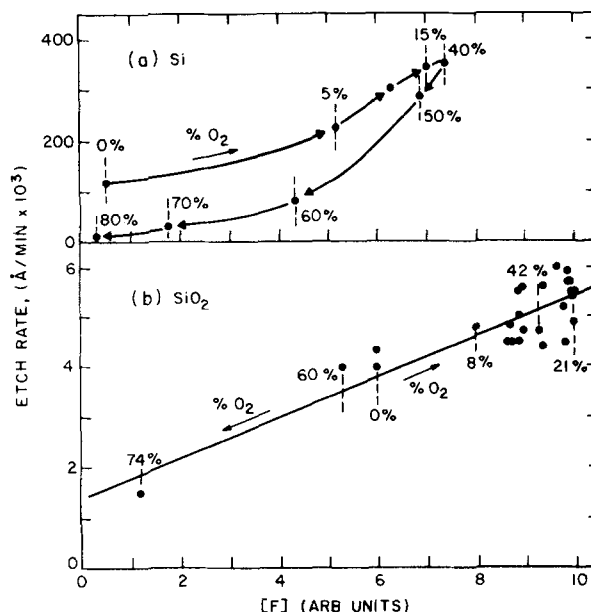


FIG. 6. Etch rates vs F atom concentration as measured by emission intensity: (a) silicon etching, arrows show direction of oxygen increase in the feed; (b) SiO_2 etching, straight line determined from a linear regression of experimental data.

in the level of oxygen atoms [Fig. 4(b)], while SiO_2 samples of similar area had no perceptible influence on the emission intensities. As oxygen is added to the feed, the intensity of F-atom emission first increases, goes through a maximum, and then decreases. This behavior has also been found when CF_4 is mixed with oxygen in a discharge,^{1c,d,g,h} and $\text{CF}_4\text{-O}_2$ data for the present discharge conditions are included in Fig. 4(a) for comparison (maximum $\text{CF}_4\text{-O}_2$ emission is normalized to SF_6 results; absolute intensity is $\approx \frac{1}{15}$ F emission from SF_6). Atomic oxygen emission [Fig. 4(b)] also increases with the fraction of O_2 in the feed, in agreement with previous^{1c,d,g} results for $\text{CF}_4\text{-O}_2$ plasmas.

Since the population of excited states depends on electron energy and density, which in turn may vary with composition (at constant power and pressure), the concentrations of F and O atoms are not necessarily a linear function of emission intensity. A modest deviation from linearity has been reported for $\text{CF}_4\text{-O}_2$ discharges,^{1d,g} but the effect will be less important in $\text{SF}_6\text{-O}_2$ because electron energies at moderate to high E/p in SF_6 ,^{11a} O_2 ,¹⁹ and atomic oxygen²⁰ are very similar. Thus at constant pressure and discharge conditions, F-atom emission from $\text{SF}_6\text{-O}_2$ should be nearly proportional to atomic fluorine.

The etch rates of silicon and SiO_2 in $\text{SF}_6\text{-O}_2$ are given in Fig. 5; etch rates of SiO_2 in $\text{CF}_4\text{-O}_2$ mixtures under the same conditions are included for comparison. From Figs. 4 and 5 it may be seen that the variation of etch rates is related to the concentration of atomic fluorine. Figure 6 shows these etch rates as a function of F-atom concentration (as measured by the 703.7-nm intensity).

As is evident from Fig. 6(a), a given [F] can yield two different Si etch rates depending on $[\text{O}_2]$. This hysteresis effect has previously been noted in $\text{CF}_4\text{-O}_2$ discharges under similar conditions,^{1d,g,h} and has been

quantitatively explained^{14,6} in terms of the competition of oxygen atoms with fluorine atoms for chemisorption on the silicon surface. The existence of this same effect in a different chemistry, the SF₆-O₂ system, lends additional support to the theory: for a given F-atom concentration, increasing oxygen in the feed results in a greater oxygen coverage and a lower etch rate. It may also be observed that "inversion" of the Si etch rate begins at $x_{O_2} \sim 0.4$ [Fig. 4(b)], the point at which O production increases most rapidly and where the introduction of a Si surface first leads to a noticeable decrease in the concentration of atomic oxygen.

The SiO₂ etch rates are given in Fig. 6(b). As with CF₄-O₂ discharges, O₂ has no retarding influence on this etch rate. On the other hand, the positive intercept (Fig. 6) suggests that sulfuryl radicals or ion bombardment may contribute to etching when the concentration of fluorine is low. Similar results have been reported for CF₄-O₂ discharges and SF₆ discharges at higher pressure.⁵ In this latter work, it was suggested that SF₃ and/or SOF₃ could etch SiO₂.

Finally, it may be noted that etch rates in SF₆-O₂ are 5–10 times faster than those in CF₄-O₂ plasmas under similar conditions; the etch rate selectivity of Si:SiO₂ is also high. Figure 6 shows the relative etch rates Si:SiO₂ $\sim 70:1$; however, the etching of Si is exothermic and the temperature of the Si substrates during etching may have increased relative to SiO₂. Flamm *et al.*²¹ recently showed that the "intrinsic" etch ratio of Si:SiO₂ by F atoms ranges from $\sim 41:1$ at room temperature to $26:1$ at 100 °C; considering the possible temperature increase, the present data are consistent with F-atom etching rates.

Several patterned Si samples were also etched using 10% O₂-SF₆ in a 20-cm-diam radial-flow parallel-plate reactor (0.3 Torr, 50 sccm, 50 W, 2.5-cm plate spacing, 23 °C electrode). Rapid silicon etching (~ 3500 Å/min) with high selectivity (Si:SiO₂ 31:1) was obtained and, qualitatively, the etched surfaces appeared to be smoother than those obtained in CF₄-O₂ mixtures.

CONCLUSIONS

SF₆-O₂ mixtures may represent an attractive alternative to CF₄-O₂ for the plasma etching of silicon and SiO₂. Rapid etch rates can be obtained along with a high selectivity in favor of silicon. The present data support the chemisorption model of hysteresis^{14,6} during the etching of Si in plasmas containing both oxygen and fluorine atoms. The reaction kinetics and products of SF₆-O₂ plasmas are generally analogous to CF₄-O₂ plasmas¹⁴—the principal differences in SF₆-O₂ being the dominance of dissociative attachment among electron impact processes, much larger concentrations of atomic fluorine, and the reaction of SF₆ to form oxyfluorides, fluorine, and SiF₄ as sole final products. More work is necessary to further clarify these mechanisms and to establish optimal processing parameters for practical etching.

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- ¹(a) H. Abe, in Proceedings of the Third International Symposium on Plasma Chemistry, University of Limoges, Limoges, France, 1977 (unpublished), p. RT-5; (b) C. J. Mogab, J. Electrochem. Soc. 124, 1262 (1977); (c) W. R. Harshbarger, T. A. Miller, P. Norton, and R. A. Porter, Appl. Spectrosc. 31, 201 (1977); (d) C. J. Mogab, A. C. Adams, and D. L. Flamm, J. Appl. Phys. 45, 3796 (1978); (e) H. F. Winters, J. W. Coburn, and E. Kay, *ibid.* 48, 4973 (1977); (f) J. W. Coburn and E. Kay, Solid State Technol. 22, 117 (1979); (g) D. L. Flamm, *ibid.* 22, 109 (1979); (h) G. Smolinsky and D. L. Flamm, J. Appl. Phys. 50, 4982 (1979).
- ²(a) H. Boyd and M. S. Tang, Solid State Technol. 22, 133 (1979); (b) Matheson Tech Brief No. TB-151 (unpublished).
- ³R. Heineke, Standard Telecommunications Laboratory CVD Project RP7-78 Annual Report, 1972 (unpublished).
- ⁴J. J. Wagner and W. W. Brandt, in Proceedings of the Fourth International Symposium on Plasma Chemistry, Zurich, 1979 (unpublished), p. 120.
- ⁵(a) G. Bruno, P. Capezzuto, F. Cramarossa, R. d'Agostino, and G. Latrofa, in Ref. 4, p. 460; (b) G. Bruno, R. Capezzuto, F. Cramarossa, R. d'Agostino, and E. Molinari, J. Fluorine Chem. 16, 209 (1980).
- ⁶(a) D. Edelson, C. A. Bieling, and G. T. Kohman, Ind. Eng. Chem. 45, 2094 (1953); (b) H. J. Emeleus and B. Tittle, J. Chem. Soc. 1644 (1963); (c) D. C. Frost, S. T. Lee, C. A. McDowell, and N. P. C. Westwood, J. Electron Spectrosc. Relat. Phenom. 12, 95 (1977); (d) L. G. Christophorou, D. R. James, R. Y. Pai, R. A. Mathis, M. O. Pace, D. W. Bouldin, A. A. Christodoulides, and C. C. Chan, Oak Ridge National Laboratory Semiannual Report No. ORNL/TM-6113, 1977 (unpublished).
- ⁷M. Schmidt and W. Siebert, in *Sulfur*, p. 843, in *Comprehensive Inorganic Chemistry*, edited by A. F. Trotman-Dickenson (Pergamon, New York, 1973), p. 795.
- ⁸(a) W. H. Beattie, G. P. Arnold, and R. G. Wenzel, Chem. Phys. Lett. 16, 164 (1972); (b) J. L. Lyman, Appl. Opt. 12, 2736 (1973).
- ⁹P. C. Nordine and D. E. Rosner, J. Chem. Soc., Faraday Trans. 1 72, 1526 (1976); P. C. Nordine and J. D. LeGrange, AIAA J. 14, 644 (1976).
- ¹⁰G. Bruno, P. Capezzuto, and F. Cramarossa, J. Fluorine Chem. 14, 115 (1979).
- ¹¹(a) L. E. Kline, D. K. Davies, C. L. Chen, and P. J. Chantry, J. Appl. Phys. 50, 6789 (1979); T. Yoshizawa, Y. Sakai, and H. Tagashira, J. Phys. D 12, 1839 (1979); (b) R. E. Kennerly, R. A. Bonham, and M. McMillan, J. Chem. Phys. 70, 2039 (1979); B. Lehman, Z. Naturforsch. A 25, 1755 (1970); (c) D. C. Frost and C. A. McDowell, J. Chem. Phys. 29, 503 (1958); W. M. Hickam and R. E. Fox, J. Chem. Phys. 25, 642 (1956); S. Avrillier and J. P. Schermann, Opt. Commun. 19, 87 (1976); (d) C. L. Chen and P. J. Chantry, J. Chem. Phys. 71, 3897 (1979).
- ¹²C. L. Chen and P. J. Chantry, Bull. Am. Phys. Soc. 17, 406 (1972).
- ¹³J. P. McGeehan, B. C. O'Neill, A. N. Prasad, and J. D. Craggs, J. Phys. D 8, 153 (1975).
- ¹⁴S. W. Benson and J. Bott, Int. J. Chem. Kinet. 1, 451 (1969).
- ¹⁵R. Hofland, M. L. Lundquist, A. Ching, and J. S. Whittier, J. Appl. Phys. 45, 2207 (1974); H. Chen, R. E. Center, D. W. Trainor, and W. I. Fyfe, *ibid.* 48, 2297 (1977); A. Mandl, B. Kivel, and E. W. Evans, J. Chem. Phys. 53, 2363 (1970); V. H. Shui and J. C. Keck, *ibid.* 59, 5242 (1973); G. E. Caledonia, Chem. Rev. 75, 333 (1978); R. K. Boyd and G. Burns, J. Phys. Chem. 83, 88 (1979); C. J. Ultee, Chem. Phys. Lett. 46, 366 (1977); P. S. Ganguli and M. Kaufman, *ibid.* 25, 221 (1974); W. D. Breshears and P. F. Bird, J. Phys. Chem. 58, 5176 (1973); B. I. Schneider

- and C. A. Brau, Appl. Phys. Lett. 33, 569 (1978); R. J. Hall, J. Chem. Phys. 68, 1803 (1978); W. L. Nighan, Appl. Phys. Lett. 32, 297 (1978); W. Tam and S. F. Wong, J. Chem. Phys. 68, 5626 (1978); D. W. Trainor and J. H. Jacob, Appl. Phys. Lett. 35, 920 (1979).
- ¹⁶K. L. Wray and E. V. Feldman, in Proceedings of the Fourteenth International Symposium on Combustion (The Combustion Institute, Pittsburgh, Pa., 1973), pp. 229-38.
- ¹⁷B. Siegel and P. Breisacher, J. Inorg. Nucl. Chem. 31, 675 (1969).
- ¹⁸JANAF Thermochemical Tables, 2nd ed., NSRDS-NBS Publication 37, edited by D. R. Stull and H. Prophet (U. S. GPO, Washington, D.C., 1971); M. W. Chase, J. L. Curnutt, A. T. Hu, H. Prophet, A. N. Syverud, and L. C. Walker, J. Phys. Chem. Ref. Data 3, 311 (1974); M. W. Chase, J. L. Curnutt, R. A. McDonald, and A. N. Syverud, *ibid.* 7, 793 (1978); R. L. Wilkins, Aerospace Corp. Report No. TR-0158 (3240-20)-19, 1968 (unpublished).
- ¹⁹K. Masek, T. Ruzick, and L. Laska, Czech. J. Phys. B27, 888 (1977).
- ²⁰M. Capitelli, M. Dilonardo, and C. Gorse, Beitr. Plasma-phys. (to be published).
- ²¹D. L. Flamm, V. M. Donnelly, and J. A. Mucha, J. Appl. Phys. (to be published).